SMART VALVE OVERVIEW

Introducing the Smart Valve™: An Innovative Solution for Optimized Water Savings

What is the Smart Valve™?

The Smart Valve™ is a US patented device designed to optimize water flow and consumption, ensuring accurate meter readings and significant cost savings. Here's how it works:

- **Air Compression:** Maintains city pressure upstream of the water meter to prevent air volume expansion, ensuring accurate meter readings.
- **Flow Stabilization:** Uses variable adjusting technology to create a smooth, even flow, protecting against pressure surges.
- **Reduced Consumption:** Creates a back pressure upstream resulting in a downstream pressure drop, reducing actual water consumption in non-volumetric uses.
- **Pressure Reduction:** Achieves significant cant pressure reduction in high-PSI water systems, minimizing issues from excessive water pressure.
- **Secondary Backflow Prevention:** Acts as a secondary back flow preventer, protecting the municipal water supply from contamination.

How Does the Smart Valve™ Save on Water?

- **Elimination of Air Volume Charging:** Compresses air before the meter to ensure only actual water volume is measured.
- **Reduction in Actual Water Consumption:** Controls pressure drops downstream, optimizing flow in showers and sinks.
- **Protection from Pressure Surges:** Smoothens water flow, preventing plumbing damage and waste.
- **Reducing Excessive Water Pressure:** Minimizes over-consumption and reduces wear and tear on fixtures.
- **Conservation of Water:** Helps in water conservation, especially in facilities without pressure reducers or regulators.

Peter Chochua, VP Major Projects & Partnerships
Peter.chochua@americanwatersavings.com

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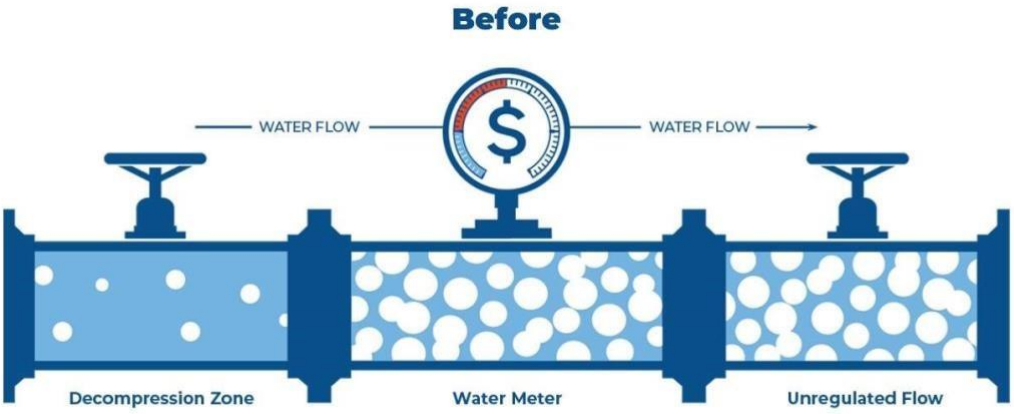
Smart Valve Summary

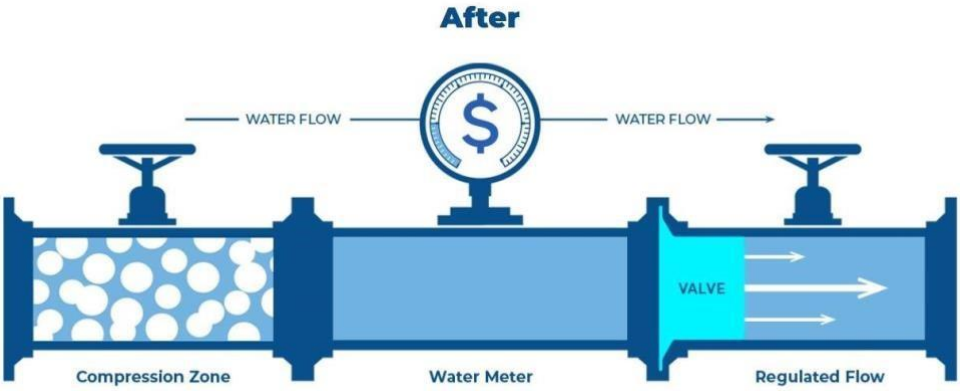
*One of the most common issues when it comes to water consumption is the **charging for volume of air** in the water as it is delivered to your site from the municipality. **Essentially 99% of water meters in North America charge you for air** inside of the water as it travels through your meter.*



Based on the laws of physics and fluid dynamics, the Smart Valve utilizes hydro-static back pressure to create a compression zone underneath the meter, upstream from the point of installation.

Standard Water Meters are designed to measure total volume but can't tell the difference between liquid & gas. By creating a compression zone, the air has no physical space to expand as it passes through your meter.

Without any area to expand, the air remains compressed, and the meter can accurately register the volume of actual water passing through, leading to a significant reduction in usage and costs.



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AMERICAN WATER SAVINGS INC.

Smart Valve Technical and Safety Details

Installation and Compatibility

- **Easy Installation:** The Smart Valve™ is externally adjustable and is installed on the client side of the water meter. The installation process requires a water shutdown for up to 3.5 hours. This is the only time we will ask for a water shut down.

- **Calibration:** Simple fingertip calibration after installation.

- **Compatibility:** Fits 99% of all water meters in North America and comes in various sizes and fittings to accommodate different pipe dimensions.

Maintenance and Lifespan

- **No Maintenance Required:** Designed to work 24/7/365 without the need for maintenance.
- **Durable Construction:** Made with high-quality materials and backed by a 10-year manufacturer's warranty.

Safety and Performance

- **Prevents Pressure Surges:** Smoothens water flow to prevent leaks and damage.
- **High Pressure Handling:** Reduces high water pressure without compromising performance.
- **Material:** Constructed from Acetron GP, holding NSF 61 and NSF 372 certifications.
- **No External Power:** Operates using the flow and pressure of the water system, requiring no batteries or external power sources.

Additional Features

- **Manual Adjustments:** Can be manually adjusted externally after installation without the need of a water shut down.
- **Automatic Adjustments:** Once calibrated, automatically adjusts to changes in water flow and pressure.
- **Protective Design:** Stabilizes flow, reduces pressure surges, and controls water consumption to protect existing infrastructure.

Peter Chochua, VP Major Projects & Partnerships
Peter.chochua@americanwatersavings.com

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Smart Valve FAQ

Q How is the Smart Valve Installed?

One-time Installation: The Smart Valve™ is externally adjustable and is installed once on the client side of the water meter. It doesn't require turning off the water or removal of the device for adjustments.

Calibration: Once installed and the water is turned back on, the Smart Valve™ is calibrated to the perfect setting for the system. Calibration is easy and can be done with the strength of a fingertip.

Q Is the Smart Valve compatible with my current system?

Adaptable Design: The Smart Valve™ is designed to be compatible with a wide range of systems as it can be adjusted to various settings and can adapt to changes in system pressure or flow rate. It is compatible with 99% of all water meters in North America.

Various Sizes and Fittings: The Smart Valve™ comes in different sizes and fittings (double flanged or female threaded) to accommodate different pipe dimensions, ensuring compatibility with most water systems.

Q What kind of maintenance does the Smart Valve™ require?

No Maintenance Required: The Smart Valve™ is designed to work 24/7/365 without the need for maintenance, making it a "set-it-and-forget-it" solution.

Durable and Reliable: Made with high-quality materials and precision manufacturing, the Smart Valve™ is built to last, further reducing the need for maintenance.

Q What is the lifespan of the Smart Valve?

Durable Construction: The Smart Valve™ is made with high-quality materials, ensuring a long lifespan, and is backed by a 10-year manufacturer's warranty.

Q Can the Smart Valve handle high pressure and flow rates?

High Pressure Handling: The Smart Valve™ can reduce high water pressure in systems without compromising performance.

Accommodates High Flow Rates: The Smart Valve™ design allows it to work efficiently even in systems with high flow rates.

Q What is the Valve made of?

Made from Acetron GP: The Smart Valve™ is constructed using Acetron GP, a material known for its strength and durability, making it ideal for long-term performance in various water systems.

Certified Construction: With its Acetron GP construction, the Smart Valve™ also holds NSF 61 and NSF 372 certifications, ensuring compliance with material safety and performance standards.

Peter Chochua, VP Partnerships
Peter.chochua@americanwatersavings.com

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Q Where does the air go?

Compression and Passage: The Smart Valve™ compresses the air mixed with water and allows it to pass through the water meter in a compressed state. This ensures the meter measures the actual water volume and not the volume of air.

Return to Original State: After passing through the Smart Valve™, the air returns to its original lower pressure state downstream, without affecting the water pressure entering the facility.

Q Does the installation of the Smart Valve require any major changes to my existing water system?

No Major Changes Required: The Smart Valve™ is designed for easy external installation on the existing water line and does not require any major changes to the existing piping system.

Q How is the Smart Valve powered? Does it require batteries or external power?

No External Power Required: The Smart Valve™ operates using the flow and pressure of the water in the system and does not require batteries or a connection to the mains for power.

Self-Operating: It is a mechanical device that automatically adjusts to the water ow and pressure, making it self-operating without the need for any external power source.

Q Can the valve be controlled and adjusted manually?

Externally Adjustable: The Smart Valve™ can be manually adjusted externally after installation to calibrate it to the perfect setting for your system.

Automatic Adjustments: Once calibrated, the Smart Valve™ automatically adjusts to changes in water flow and pressure, ensuring optimal performance without the need for constant manual intervention.

Q Is there any risk that the Smart Valve can cause damage to my water system?

Designed for Safety: The Smart Valve™ is designed with safety in mind and aims to prevent damage to existing infrastructure.

Acts as a Protective Measure: By stabilizing flow, reducing pressure surges, and controlling water consumption, the Smart Valve™ protects the existing infrastructure and minimize the risk of damage that could be caused by excessive pressure or unpredictable water flow.

Peter Chochua, VP Partnerships
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Smart Valve Program Summary

Guaranteed 15% Savings: Contractually backed, minimum savings on cost and consumption.

- Shared Savings Program split typically yields a NET 10% savings, with 5% paid for the program.
- The Smart Valves and all servicing are provided as part of the Shared Savings Program.
- Installation fees apply per Smart Valve installed.
- Monthly shared savings program costs are paid via pre-approved ACH Withdrawal or Credit Card.

Guaranteed Savings? What happens if it doesn't hit the mark?

The Shared Savings Program is backed by a 6-month trial period, during which, if you do not attain at least a 15% reduction in metered water consumption against your baseline, you may opt for one of two courses of action:

- Uninstall the Smart Valves (at our expense) and be reimbursed ALL associated costs (Installation & monthly payments).
- Split the difference based on the results being attained, reflected in the Shared Savings Program costs. For instance, if you are attaining a 12% reduction, we will reduce our Shared Savings Program cost by half of the difference (1.5%) to 3.5%. If you only attain a 10% reduction, the Shared Savings Program cost is reduced to 2.5%.

Peter Chochua, VP Partnerships
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American Water Savings Process

Step 1: Provide 12 months of usage data (water bills) for each potential site where you would like to install the Smart Valve.

Step 2: A proposal is generated to present an accurate projected savings figure with the current baseline of water consumption and the analytics of our program.

Step 3: Upon approval, we will send you an executable agreement detailing the 6-month money-back guarantee period, installation process with terms & conditions.

Step 4: Upon receiving signed agreement, we will schedule a site assessment to qualify the property. (This does not require any water shutoff)

Please note, If the site is deemed not eligible, it does not qualify for our program, and we will not proceed with the installation. (site assessments do not incur any costs)

Step 5: Installation has been successfully completed; we will return within 4 weeks to start the calibration phase.

Typically, we want to calibrate the valve in phases, this ensures that we match the specific pressure profile of your building. (This will avoid any change in pressure issues for your tenants.)

Ongoing: Once the calibration phase has been completed, we will closely monitor the usage data. Every 6-12 months, our team will complete a comprehensive Measurement & Verification Process to verify the savings.

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Notable Clients and References

We are proud to work with some of the most recognized brands across the country.

Other industry specific references are available upon request

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